

t large enough to capture the carries. But we can compute the condition for $v=1$, $v=2$, $v \geq 3$, etc. by looking at 3

Let's compute for each r :

$r=1$: $3 \cdot 1 + 1 = 4$. So $3x+1 = 48a + 4 = 4(12a+1)$. So $v = 2 + v_2(12a+1)$. $12a+1 \bmod 2$: $12a$ is even, so $12a+1$ is odd.

$r=3$: $3 \cdot 3 + 1 = 10$. $3x+1 = 48a+10 = 2(24a+5)$. $24a+5$ is odd? $24a$ even, $+5$ odd $\Rightarrow$ odd. So $v=1$ always. $k=1$ always.

$r=5$: $3 \cdot 5 + 1 = 16$. $3x+1 = 48a+16 = 16(3a+1)$. So $v = 4 + v_2(3a+1)$. $3a+1 \bmod 2$: if a even, $3a$ even, $+1$ odd $\Rightarrow v_2 \geq 1$.

$r=7$: $3 \cdot 7 + 1 = 22$. $3x+1 = 48a+22 = 2(24a+11)$. $24a+11$ odd? $24a$ even, 11 odd $\Rightarrow$ odd. So $v=1$ always. $k=1$. $T(x)=1$.

$r=9$: $3 \cdot 9 + 1 = 28$. $3x+1 = 48a+28 = 4(12a+7)$. So $v=2+v_2(12a+7)$. $12a+7$: $12a$ even, 7 odd $\Rightarrow$ odd? Even+odd=odd.

$r=11$: $3 \cdot 11 + 1 = 34$. $3x+1 = 48a+34 = 2(24a+17)$. $24a+17$ odd? $24a$ even, 17 odd $\Rightarrow$ odd. $v=1$. $k=1$. $T=24a+17 \bmod 16$.

$r=13$: $3 \cdot 13 + 1 = 40$. $3x+1 = 48a+40 = 8(6a+5)$. So $v=3+v_2(6a+5)$. $6a+5$: if a even, $6a$ even, 5 odd $\Rightarrow$ odd $\Rightarrow v_2 \geq 0$.

$r=15$: $3 \cdot 15 + 1 = 46$. $3x+1 = 48a+46 = 2(24a+23)$. $24a+23$ odd? even+odd=odd. $v=1$. $k=1$. $T=24a+23 \bmod 16$: $24a \bmod 16 = 8a \bmod 16$.

Summary table for residues mod 16:

$r=1$: $k=2$, next in $\{1,5,9,13\}$

$r=3$: $k=1$, next in $\{5,13\}$

$r=5$: $k \geq 4$, next any odd (since $v=4$ gives all, deeper also possible)

$r=7$: $k=1$, next in $\{11,3\}$

$r=9$: $k=2$, next in $\{7,3,15,11\}$

$r=11$: $k=1$, next in $\{1,9\}$

$r=13$: $k=3$, next any odd

$r=15$: $k=1$, next in $\{7,15\}$

Now we want to analyze the maximum deviation of sum k from $2m$ along paths. Since we can't have negative k ,

- From 3 ($k=1$): goes to 5 or 13. 5 has high k (≥ 4), 13 has $k=3$. So leaving 3 always leads to a state with $k \geq 3$.

- From 7 ($k=1$): goes to 11 or 3. 11 has $k=1$, 3 has $k=1$. So from 7 $\rightarrow$ 11 $\rightarrow$ 1 or 9? 11 goes to 1 or 9 ($k=2$). So 7 $\rightarrow$ 1 or 9 ($k=2$).

- From 11 ($k=1$): goes to 1 or 9 ($k=2$). So one $k=1$ then $k=2$.

- From 15 ($k=1$): goes to 7 or 15. $15 \rightarrow 7 \rightarrow \dots$ as above, or $15 \rightarrow 15$ (self-loop?). $15 \rightarrow 7$ or 15 . If $15 \rightarrow 15$, that's $k=1$.

Now, to get low sum k , we might try to alternate between $k=1$ and $k=2$, etc. But we can compute the minimum k for any state.

Maybe we can prove that for any long path, the sum of k is at least $2m - O(\log \max)$. We could use potential function.

Perhaps the right approach is to consider the "2-adic" valuation of the numbers and show that the cumulative sum of k is bounded.

For now, I'll present the constructed graph and the insight that state 15 can have long $k=1$ chains but limited by v_2 .